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CMU-Africa

April 26, 2026



Day reference



Night query

Problem

We want a robot to recognize its location at night using a daytime map. This is hard because large illumination changes make many daytime-trained descriptors unstable at night.

Gap in existing work

Many VPR papers report strong benchmark performance, but it is still unclear which descriptors **transfer best to our specific setting**: day-to-night localization, training-free evaluation, and a small campus deployment scenario.

Research question

If we keep the pipeline simple and avoid retraining, which descriptor gives the most reliable nocturnal place recognition on both a controlled benchmark and our own campus dataset?

Hypothesis

Descriptor choice is the dominant factor in our current nocturnal VPR setup. We expect modern VPR descriptors to outperform older or more generic baselines on both GardensPoint and the strict campus dataset.

Descriptors evaluated

- **Older / generic:** SAD, AlexNet, NetVLAD
- **Local + global:** PatchNetVLAD
- **Modern VPR descriptors:** CosPlace, EigenPlaces, SALAD, SelaVPR++
- **Specialized spiking model:** VPRTempo

What these are

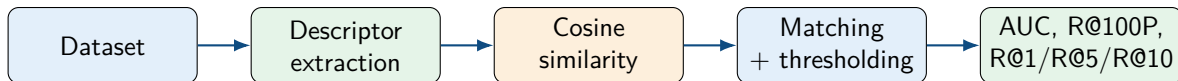
CosPlace / EigenPlaces learn strong global place descriptors. SALAD improves descriptor aggregation with DINOv2 features. SelaVPR++ adapts foundation models for efficient place recognition. VPRTempo is a fast temporally encoded spiking network rather than a universal drop-in descriptor.

Why this comparison matters

Our current phase focuses on **descriptor exploration first**: before adding CLAHE, sequence matching, or temporal filtering, we want to know which representation is strongest in our actual setting.

Methods still planned

CLAHE preprocessing, sequence matching, and temporal prediction remain future or next-phase improvements once the descriptor baseline is settled.



GardensPoint

200 day-right reference images and 200 night-right queries. This remains our main benchmark because it directly matches the **day-to-night** problem.

Campus dataset

50 day images and 64 night queries. In this talk we focus on the **strict image-level protocol**, which is the more conservative evaluation. All experiments are **training-free transfer tests** unless explicitly stated otherwise.

Descriptor	AUC	R@1	R@5	R@10
CosPlace	0.592	0.350	0.840	0.925
EigenPlaces	0.676	0.370	0.840	0.940
PatchNetVLAD	0.802	0.440	0.800	0.840
SALAD	0.924	0.450	0.955	0.990
SelaVPR++	0.937	0.465	0.950	0.995
VPRTempo	0.065	0.095	0.230	0.290

Takeaway

The strongest benchmark descriptors are now **SelaVPR++** and **SALAD**. VPRTempo transfers very poorly with the official demo checkpoint.

Strict interpretation

No obvious leakage was found, but the most trustworthy

Examples: 1 correct (green) and 1 wrong (red) matches from $S \geq \text{thresh}$



Why we keep it

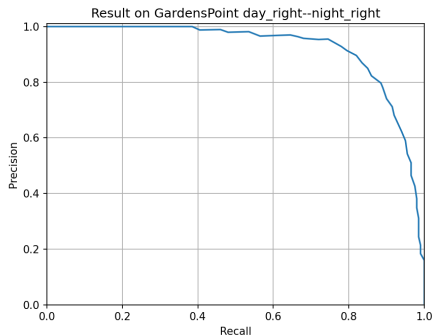
GardensPoint is still the right primary benchmark for this project because it directly matches the task we care about: **day-to-night place recognition**. It gives us a clean, controlled test before moving to the noisier campus setting.

What we learn from it

- whether a descriptor can handle a strong day-night appearance gap
- whether the correct place is ranked first or only appears later in the list
- whether a strong result is consistent across many query frames

How we interpret it

We treat GardensPoint as the **controlled benchmark**, then



Descriptor	AUC	R@100P	R@1	R@5	R@
CosPlace	0.470	0.030	0.727	1.000	1.0
HDC-DELF	0.488	0.030	0.788	0.939	0.9
SALAD	0.504	0.030	0.818	1.000	1.0
SelaVPR++	0.464	0.061	0.909	1.000	1.0
VPRTempo	0.064	0.000	0.182	0.515	0.6

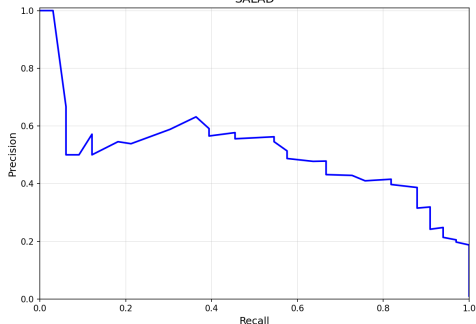
Interpretation

Under strict 1-to-1 labels, **SelaVPR++ gives the strongest top-1 retrieval** while SALAD stays strong. No-match rejection remains difficult, and VPRTempo is not competitive in this transfer setting.

Examples: 3 correct (green) and 5 wrong (red) matches from $S \geq \text{thresh}$



Precision-Recall Curve - Campus Dataset
SALAD



Key observation

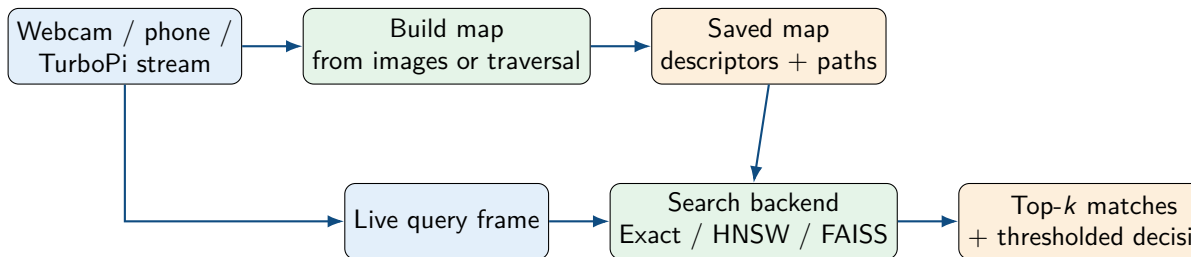
The campus setting is much harsher than GardensPoint because it mixes day-night change with viewpoint change, limited map coverage, and many hard no-match cases.

Interpretation

- **SelaVPR++** gives the best top-1 retrieval under the strict protocol.
- **SALAD** stays consistently strong and reaches perfect R@5 and R@10.
- **CosPlace** remains a strong practical baseline.
- **VPRTempo** is not competitive in this transfer setup.

Main lesson

Strong modern descriptors clearly help, but descriptor quality alone does not fully solve rejection and ambiguity in a small real campus deployment.



Current status

We already have an end-to-end live pipeline for map building and localization. The next step is to connect the strongest descriptor choices and lightweight temporal ideas to the demo path.

- 1 Our hypothesis is supported: descriptor choice strongly affects nocturnal VPR performance.
- 2 GardensPoint remains the right primary benchmark because it directly matches the day-to-night problem.
- 3 The best descriptors we tested are now **SelaVPR++** and **SALAD**, with CosPlace still a strong practical baseline.
- 4 The strict campus dataset is much harder than GardensPoint and exposes the remaining gap between benchmark success and deployment success.
- 5 VPRTempo did **not** transfer well here, mainly because the official checkpoint is a Nordland demo model rather than a universal descriptor.
- 6 The live system is already in place, so the next phase is to connect the strongest descriptors and lightweight temporal improvements to a robust demo.

What we would do next

- integrate **SALAD** and **SelaVPR++** into the live path as the strongest current descriptors
- test lightweight improvements such as **CLAHE** and **sequence matching**
- improve **no-match rejection** for campus deployment
- compare accuracy against runtime and deployment cost for a practical robot demo

Demo message

Our contribution is not only a benchmark comparison, but a practical descriptor study for night-time localization in a real campus environment.